

EMPIRICAL RESEARCH · BITCOIN FACTORS · CORRELATION VS. LEADING INDICATORS

What actually drives Bitcoin: a high-beta risk asset that follows global liquidity — not digital gold, and not a falling-dollar trade.

Sample 2015–2026 · first-party empirical engine + STORM cross-check ·

+29.8pp

BTC FWD-6M RETURN SPREAD
WHEN M2 IS ACCELERATING
(THE TOP LEADING INDICATOR)

0.28 → 0.52

BTC-NASDAQ DAILY
CORRELATION: FULL SAMPLE
VS. LAST 12 MONTHS (RISK-
BETA, RISING)

+0.09

BTC-GOLD FULL-SAMPLE
DAILY CORRELATION —
STATISTICALLY ~ZERO
("DIGITAL GOLD" FAILS)

Executive summary

The data overturns the lazy version of every Bitcoin narrative. Bitcoin is, first and foremost, a **high-beta risk asset whose equity correlation has structurally risen** — its daily correlation to the Nasdaq climbed from ~0.28 over the full 2015–26 sample to ~0.52 over the last year — not "digital gold" (full-sample gold correlation ~0.09, statistically indistinguishable from zero). The single most useful *leading* indicator is **global liquidity / M2 growth**: when US M2 year-over-year is accelerating, BTC's median forward-6-month return is +35.0% versus +5.2% when decelerating — a +29.8pp spread. Crucially, the factors most people *watch* — the Nasdaq, gold, the VIX, credit spreads — are **coincident correlates with no predictive lead**, not leading indicators. And the popular "weak-dollar / falling-real-rates" trade is the biggest myth: on a forward-return basis it is perverse in-sample. This note separates what moves *with* Bitcoin from what moves *before* it, and grades six canonical assumptions against the evidence. It is analysis, not advice, and contains no recommendation regarding any security or digital asset.

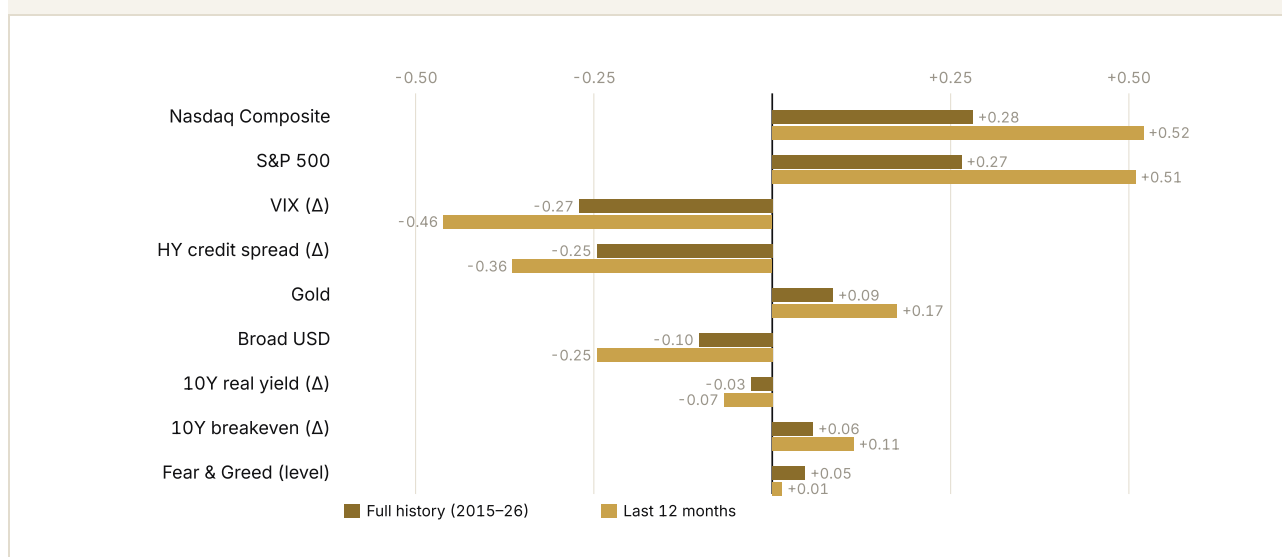
The distinction that organizes this note. *Contemporaneous correlation* is co-movement at the same time (coincident beta) — useful for hedging and risk, useless for timing. A *leading indicator* has predictive power with a lead (factor today → BTC return later) — that is what an allocator can act on. Conflating the two is the most common analytical

error in this field: a factor can be highly correlated yet have zero predictive value (the Nasdaq), or weakly correlated day-to-day yet a strong forward signal (M2).

1 · Contemporaneous correlation — and its instability

Full-sample daily-return correlations are mostly modest; the story is in the **windows**. Most factors are near-zero over the full sample but strongly correlated over the last 12 months — Bitcoin's correlation structure has shifted hard toward risk-asset behavior in the post-spot-ETF era.

EXHIBIT 1 · DAILY-RETURN CORRELATION TO BTC — FULL HISTORY VS. LAST 12 MONTHS



Pearson correlation of daily log returns (yields/spreads/vol in first differences). Source: TON618 factor engine, 2026-06-18 17:17 UTC. M2 and Fed-balance-sheet contemporaneous figures are YoY autocorrelation artifacts and are assessed as leading indicators below, not here.

2 · Coincident vs. leading — the critical split

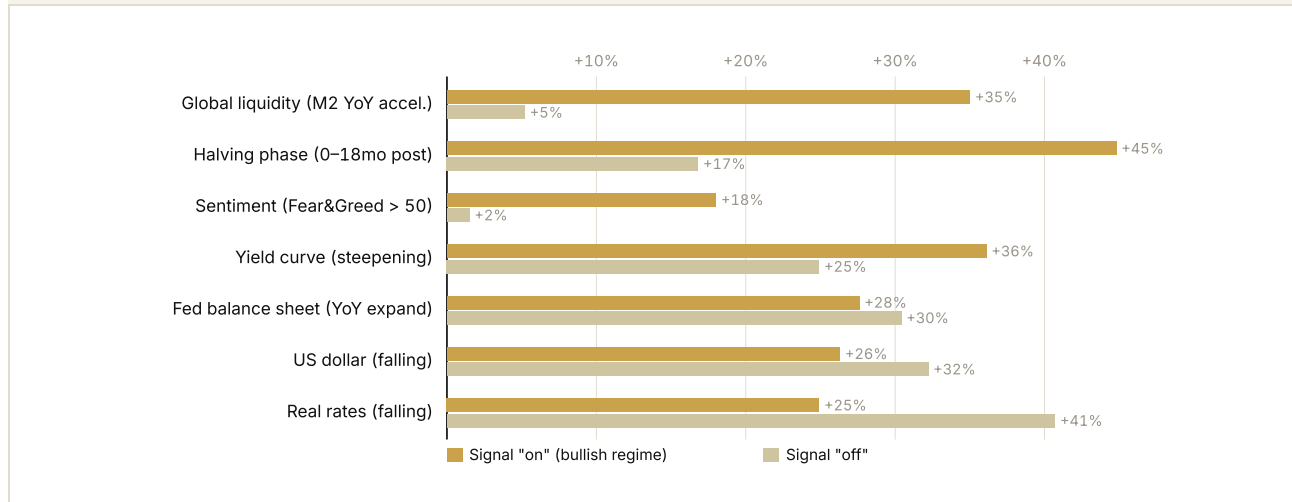
Lead/lag cross-correlation (the lag at which each factor's correlation to BTC peaks) delivers the report's sharpest result:

- **Coincident — correlates, NOT predictors** (peak at lag 0): Nasdaq, S&P 500, VIX, HY credit spreads, gold, the broad dollar. *Everything the market watches as a "Bitcoin tell" moves with Bitcoin, not before it.*
- **Lagging** (Bitcoin leads them): Treasury yields, real yields, transaction count.
- **Leading — genuine forward edge**: global liquidity / M2, and the sentiment regime. These are few, and they are the only factors with timing value.

3 · Leading-indicator ranking — forward-6-month regime split

Median BTC forward - 6 - month return when each signal is "on" (bullish regime) versus "off," monthly 2015+:

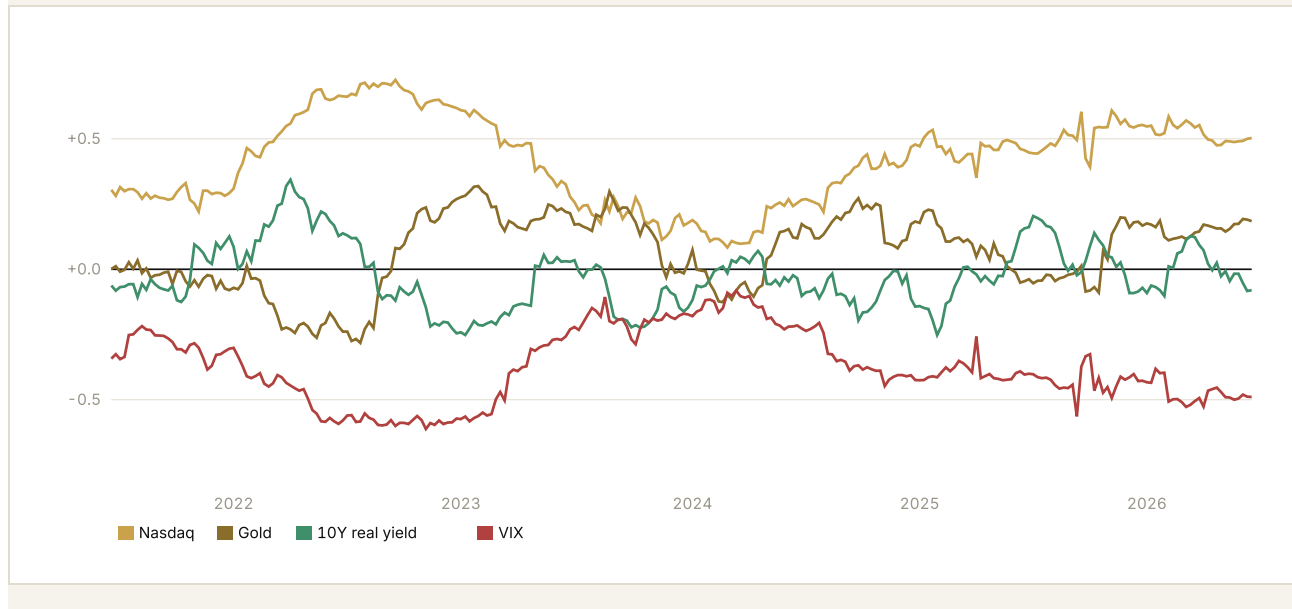
EXHIBIT 2 · MEDIAN BTC FORWARD - 6 - MONTH RETURN BY SIGNAL STATE



Source: TON618 factor engine + in-house halving model, 2026-06-18 17:17 UTC. The dollar and real-rate signals are the **wrong sign** versus popular belief — in-sample, BTC delivered higher forward returns when the dollar was firm and real rates were rising (the 2023-24 high-rate bull dominates). Fed-balance-sheet YoY carries no timing edge despite the "QE drives Bitcoin" trope.

4 · No correlation is stable

EXHIBIT 3 · ROLLING 90-DAY CORRELATION OF DAILY RETURNS (5 YEARS)



Source: TON618 factor engine, 2026-06-18 17:17 UTC. BTC-Nasdaq has ratcheted toward +0.5; BTC-real-yield turned sharply negative only recently; BTC-gold is positive but unstable. The instability is itself the most robust finding — any single-number correlation is regime-conditional.

5 • The assumption scorecard

Six canonical beliefs, graded against the first-party evidence:

COMMON ASSUMPTION	VERDICT	EMPIRICAL BASIS
"Bitcoin is digital gold / an inflation hedge"	Mostly FALSE	Full-sample BTC-gold daily corr +0.09 (≈0); gold is coincident; 10Y-breakeven corr ≈0. A 2024-26 uptick to +0.31 (90-day) exists but is small and unproven.
"Bitcoin is a risk-on Nasdaq / tech-beta asset"	TRUE — strengthening	Daily corr 0.28 (full) → 0.52 (1y); VIX -0.46, HY -0.36 over 1y; correlation rises in risk-off. The dominant contemporaneous relationship.
"Bitcoin is driven by global liquidity / M2, with a lead"	TRUE — best leading indicator	M2-accelerating regime → +35.0% vs +5.2% fwd-6m (+29.8pp, corr +0.33); in-house overlay shows ~80-day lead. Note: Fed-balance-sheet YoY alone has no timing edge.
"The ~4-year halving cycle drives price"	FRAGILE / partly true	0-18mo post-halving → +44.8% vs +16.8% fwd-6m (+28.1pp), but the forward-12-month spread collapses to +0.7pp and the test rests on ~3 independent cycles; heavily confounded with liquidity.
"Bitcoin is a weak-dollar / falling-real-rates debasement trade"	DENIED as a predictor	Forward-return spreads are perverse: dollar-falling -5.9pp, real-rates-falling -15.8pp. Only a weak, recent <i>coincident</i> dollar effect (-0.25, 1y).
"On-chain metrics (MVRV/NUPL/SOPR) time tops & bottoms"	MIXED — useful at extremes	First-party valuation metrics auth-gated this run; network-usage on-chain is reflexive/coincident, not leading. Published backtests (3 cycles) credit MVRV/NUPL with real risk-adjusted edge at extremes — but as zones, on ~3 samples (see external corroboration).

External corroboration

Independent published evidence gathered via a STORM-method, perspective-structured research pass (Stanford OVAL method, adapted); each claim cited in Sources. Where external evidence and our first-party empirics agree, confidence rises; where they diverge, we flag it.

ASSUMPTION	WHAT THE PUBLISHED LITERATURE / PRACTITIONER EVIDENCE SHOWS
Digital gold / inflation hedge	Corroborated (fails). Peer-reviewed and practitioner work finds BTC's equity correlation rose from ~0 in 2018–20 to ~0.5 after the Jan-2024 US spot-ETF launch, and that ETF approval <i>weakened</i> its hedging/diversification properties versus traditional assets — BTC now behaves more like leveraged equity than an inflation hedge. ^{[3][4][5]}
Risk-on Nasdaq / tech beta	Strongly corroborated. Independent series put the 6-month BTC–Nasdaq correlation near 0.92 by Sept 2025, with BTC–equity correlation spiking toward record highs in 2026 risk-off episodes; ETF and institutional flows now drive a larger share of BTC volume, tightening the equity link. ^{[3][5]}
On-chain valuation metrics time tops & bottoms (MVRV/NUPL)	Partly corroborated, with caveats. A peer-reviewed backtest across three cycles (Dec-2013→Apr-2025) finds the MVRV Z-score delivers higher risk-adjusted returns than buy-and-hold and Monte-Carlo random entry, and MVRV/NUPL have flagged major tops/bottoms since 2011 — but as <i>zones</i> (they can stay extreme for weeks), on only ~3 independent cycles, with model-maturation risk. ^[6]

Methodology & data

Sample: daily, 2015-01-01 to 2026-06-17. **Primary, first-party data:** BTC price and on-chain metrics — CoinMetrics community API; macro, rates, credit, liquidity — FRED; gold and equity indices — Yahoo Finance; sentiment — alternative.me Fear & Greed. Correlation is computed on returns/changes, never raw trending levels (price → log return; yields/spreads/vol → first difference; slow liquidity → YoY). Estimators: Pearson and Spearman over four windows; lead/lag cross-correlation; rolling-90-day correlation; and a forward-return regime split (median BTC fwd-6-month return, bullish vs bearish regime). **Disclosed gaps:** on-chain valuation metrics (realized cap, MVRV) were auth-gated this run and are adjudicated via published evidence; the HY credit-spread series is available only from mid-2023; small-sample cycle claims (halving, $n \approx 3$) are flagged. Methods: in-house empirical engine (primary) plus a STORM perspective-structured qualitative cross-check (Stanford OVAL method, adapted).

Risk factors & limitations

- **Regime instability is the master risk** — every correlation is window-dependent and may invert; the post-2024 ETF era is a small sample.
- **Endogeneity** — on-chain network metrics are partly caused by price; not exogenous drivers.
- **Small-sample cycle claims** — the halving rests on ~3 independent cycles and is not statistically robust.
- **In-sample inference** — 2015–26 contains few independent macro cycles; forward returns are not guaranteed.

What we are watching

- MVRV / NUPL / SOPR cycle-timing efficacy — obtain a non-gated realized-cap source and test first-party.
- Whether the post-ETF rise in BTC–equity correlation is structural or cyclical (rolling data suggests structural).
- Whether the 2024–26 BTC–gold warming is durable or a passing co-movement.
- Global (not just US) M2 lead time — reconcile the ~80-day overlay against the noisier YoY lead/lag.
- Whether the halving signal survives once liquidity is controlled for.

Sources

1. TON618 Capital factor engine (first-party): `factor_analysis.py`, run 2026-06-18 17:17 UTC. Inputs: CoinMetrics community API (BTC price, on-chain); FRED / Federal Reserve Bank of St. Louis (macro, rates, credit, liquidity); Yahoo Finance (gold, equity indices); alternative.me (Fear & Greed).
2. TON618 Capital in-house models: `btc_halving_phase.py` (halving-cycle forward returns), `btc_m2_correlation.py` (M2 lead overlay), `btc_regime_ranking.py`.
3. Institutional adoption & correlation dynamics, arXiv 2501.09911 (2025).
4. Impact of Bitcoin ETF approval on hedging properties, arXiv 2512.12815 (2025).
5. CME Group, "Why Bitcoin's Relationship with Equities Has Changed" (2025); Nasdaq and Phemex BTC–equity correlation series (2025–26).
6. On-chain measures backtest (NUPL, MVRV Z-score, CVDD), ScienceDirect S0275531926002138 (Dec-2013→Apr-2025); corroborated by checkonchain and 21Shares cycle research.

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